

Lecture 8 (153-248)

Last lecture: y_t : observed time series

$$y_t = \beta_0 + \beta_1 \cos 2\pi f t + \beta_2 \sin 2\pi f t + \varepsilon_t$$

f : unknown frequency.

$$\text{RSS}(f) = \min_{\beta_0, \beta_1, \beta_2} \sum_{t=1}^n \left(y_t - \beta_0 - \beta_1 \cos 2\pi f t - \beta_2 \sin 2\pi f t \right)^2$$

$t = 1, \dots, n$

$$f \in [0, \frac{1}{2}]$$

① Some discretization of f in $[0, \frac{1}{2}]$

$$\left\{ 0, \frac{1}{10000}, \frac{2}{10000}, \dots, \frac{5000}{10000} \right\}$$

② For each f , compute $\text{RSS}(f)$

Total computation: order $\underbrace{\text{grid size} \times n}$

Fix f , regression

e.g. $n \sim 300000$

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \text{ on } X_f = \begin{bmatrix} 1 & \cos(2\pi f t) & \sin(2\pi f t) \\ \vdots & \vdots & \vdots \\ 1 & \cos(2\pi f t) & \sin(2\pi f t) \end{bmatrix}$$

$\text{grid size} \sim 10000$
 $\rightarrow n \times 3$

$$\hat{\beta}_f = \underbrace{\begin{pmatrix} X_f^T X_f \end{pmatrix}^{-1}}_{3 \times 3} \underbrace{X_f^T y}_f \quad O(n) \text{ computation}$$

There exists a faster alternative which lets calculation of $RSS(f)$ in $O(n \log n)$ time at the **FOURIER grid**

Fourier grid
= grid of Fourier frequencies

$$= \begin{cases} 0, \frac{1}{n}, \frac{2}{n}, \frac{3}{n}, \dots, \frac{n-1}{2n} & \text{if } n \text{ odd} \\ 0, \frac{1}{n}, \frac{2}{n}, \frac{3}{n}, \dots, \frac{n/2}{n} & \text{if } n \text{ even} \end{cases}$$

Fourier frequency: f for which $nx \cdot f$ is an integer.

$$\frac{\binom{n-1}{2}}{n}$$

$$n = 5$$

$$0, \frac{1}{5}, \frac{2}{5},$$

Restriction to Fourier frequencies is done so calculations become easy.

Calculate $RSS(f)$ when f is a Fourier frequency.

$$RSS(f) = \|y - X_f \hat{\beta}_f\|^2$$

$$0 < f < \frac{1}{2}$$

$y, X_f, \hat{\beta}_f$

$$= (y - X_f \hat{\beta}_f)^T (y - X_f \hat{\beta}_f) \quad \hat{\beta}_f = (X_f^T X_f)^{-1} X_f^T y$$

= CHECK

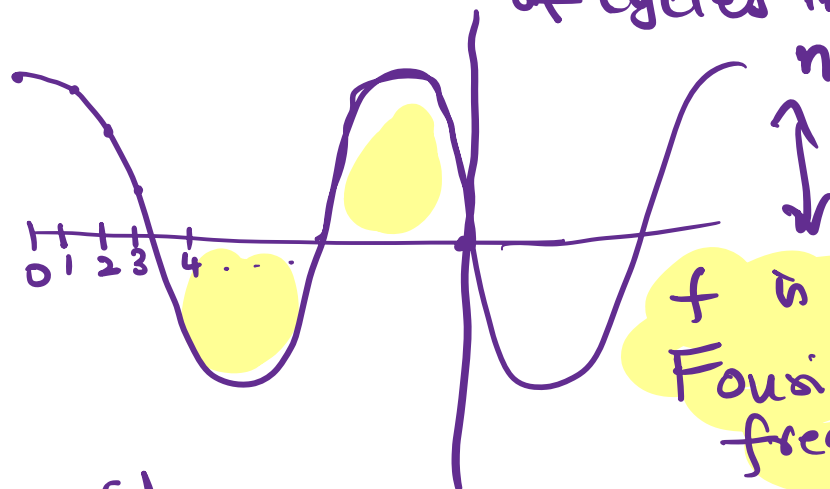
$$= y^T y - y^T X_f (X_f^T X_f)^{-1} X_f^T y$$

$$X_f^T X_f$$

$$X_f = \begin{pmatrix} 1 & \cos 2\pi f(1) & \sin 2\pi f(1) \\ \vdots & \cos 2\pi f(2) & \vdots \\ \vdots & \vdots & \vdots \\ 1 & \cos 2\pi f(n) & \sin 2\pi f(n) \end{pmatrix}$$

$$= \begin{pmatrix} \sum_{t=1}^n \cos^2 2\pi f t & \sum_{t=1}^n \cos 2\pi f t \sin 2\pi f t \\ \sum_{t=1}^n \cos 2\pi f t \sin 2\pi f t & \sum_{t=1}^n \sin^2 2\pi f t \end{pmatrix}$$

$$\sum_{t=1}^n \cos 2\pi f t = 0 \quad \text{if the cosine wave completes a whole number of cycles in time}$$



$$\sum_{t=1}^n \sin 2\pi f t = 0$$

f is a Fourier frequency

$$\sum_{t=1}^n \cos^2 2\pi f t$$

$$= \sum_{t=1}^n \left(\frac{1 + \cos 4\pi f t}{2} \right)$$

$$= \frac{n}{2} + \underbrace{\frac{1}{2} \sum_{t=1}^n \cos(4\pi f t)}_0 = \frac{n}{2}$$

$$\sum \sin^2(2\pi f t) = \frac{n}{2}$$

$$\sum (\cos 2\pi f t)(\sin 2\pi f t) = 0$$

$$X_f^T X_f = \begin{pmatrix} n & 0 & 0 \\ 0 & \frac{n}{2} & 0 \\ 0 & 0 & \frac{n}{2} \end{pmatrix}$$

→ If f is a Fourier frequency
 $0 < f < \frac{1}{2}$

$$RSS(f) = y^T y - y^T X_f (X_f^T X_f)^{-1} X_f^T y$$

$$= \begin{pmatrix} y^T X_f (X_f^T X_f)^{-1} X_f^T y \end{pmatrix} \begin{pmatrix} \frac{1}{n} \\ \frac{n}{2} \\ \frac{n}{2} \end{pmatrix} \begin{pmatrix} \sum y_t \\ \sum y_t \cos 2\pi f t \\ \sum y_t \sin 2\pi f t \end{pmatrix}$$

f : Fourier
 $0 < f < \frac{1}{2}$

$$= \frac{1}{n} \left(\sum_t y_t \right)^2 + \frac{2}{n} \left(\sum_t y_t \cos 2\pi f t \right)^2 + \frac{2}{n} \left(\sum_t y_t \sin 2\pi f t \right)^2$$

RSS(f)

$$= \underbrace{\sum_t y_t^2}_{y^T y} - \frac{1}{n} \left(\sum_t y_t \right)^2 - \frac{2}{n} \left[\left(\sum_t y_t \cos 2\pi f t \right)^2 + \left(\sum_t y_t \sin 2\pi f t \right)^2 \right]$$

$$= \sum_{t=1}^n (y_t - \bar{y})^2 - \frac{2}{n} \left[\left(\sum_t y_t \cos 2\pi f t \right)^2 + \left(\sum_t y_t \sin 2\pi f t \right)^2 \right]$$

Periodogram:

$$I(f) = \frac{1}{n} \left(\sum_t y_t \cos 2\pi f t \right)^2 + \frac{1}{n} \left(\sum_t y_t \sin 2\pi f t \right)^2$$

$$\rightarrow f \in (0, \frac{1}{2})$$

f: Fourier frequency.

$$RSS(f) = \text{variance of data} - 2 I(f)$$

for Fourier $f \in (0, \frac{1}{2})$

To compute RSS(f) efficiently, we need to compute

$$\sum_t y_t \cos 2\pi \left(\frac{j}{n} \right) t \quad \text{efficient}$$

$$\left\{ \sum_t y_t \sin 2\pi \left(\frac{j}{n}\right) t \right\} \quad j = 1, 2, \dots, \frac{n}{2}$$

$$e^{-i\theta} = \cos \theta - i \sin \theta$$

$$\rightarrow \sum_{t=1}^n y_t e^{-i 2\pi \left(\frac{j}{n}\right) t}$$

Discrete Fourier Transform (DFT)

Data: $y_0, y_1, \dots, y_{n-1}$ (Real-Valued)

DFT: $b_0, b_1, \dots, b_{n-1}$

Definition: $b_j = \sum_{t=0}^{n-1} y_t \exp(-2\pi i \frac{j}{n} t)$

FFT: FAST FOURIER TRANSFORM

→ FAST ALGORITHM For Computing the DFT.

→ np.fft.fft

Sinusoidal Model:

$$RSS(f) = \sum (y_t - \hat{y})^2 \quad \rightarrow \quad f = \frac{j}{n}$$

$$I(f)$$

$$= I\left(\frac{j}{n}\right) = \frac{|b_j|^2}{n}$$

RSS (f)

$I(f) \rightarrow$ periodogram

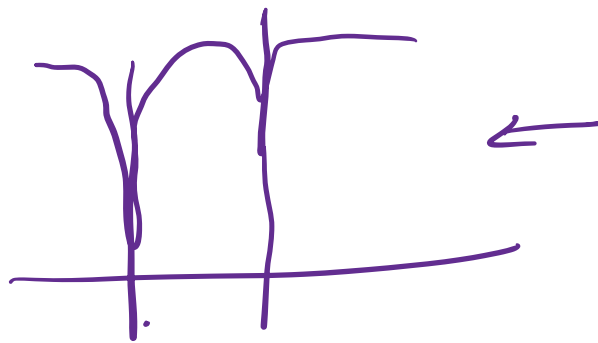
$$I\left(\frac{j}{n}\right) = \frac{|b_j|^2}{n}$$

b_j : DFT

$$y_t = \beta_0 + \beta_1 \cos 2\pi f_1 t + \beta_2 \sin 2\pi f_1 t + \beta_3 \cos 2\pi f_2 t + \beta_4 \sin 2\pi f_2 t + \varepsilon_t$$

RSS(f_1, f_2)

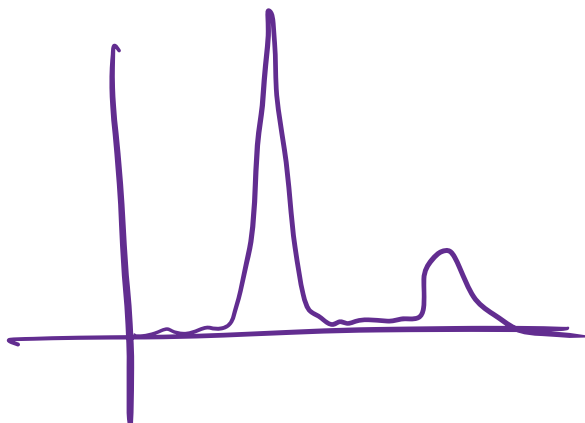
RSS(f)



Data: y_t

DFT: b_j

Periodogram: $I\left(\frac{j}{n}\right) = \frac{|b_j|^2}{n}$



Plot
A look at it
as an Exploratory
Tool.

